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An innovative TPV device design based on narrow gap IV-VI semiconductor MQW structures

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ABSTRACT

A thermophotovoltaic (TPV) device design based on multiple quantum well (MQW) materials composed of $\text{Pb}_{0.81}\text{Sn}_{0.19}\text{Se}$ wells and $\text{Pb}_{0.80}\text{Sr}_{0.20}\text{Se}$ barriers, which are nanostructured materials that can be grown on low-cost silicon, was modeled to predict electrical power generation densities. MQW materials with intersubband energy gaps of 343 meV and 450 meV in a dual junction configuration were studied. For a thermal radiator at a temperature of 1364°C the short circuit current density was estimated to be 12.1 A/cm² for each junction. Open circuit voltages for each junction ranged from 130 mV to 262 mV depending on bandgap and temperature. Power generation densities for this dual junction device increased from 2.7 W/cm² to 3.4 W/cm² as temperature decreased from 50°C to 7°C. Using a conservative value of \$1/cm² for the manufacturing of this silicon-based TPV device technology, the costs for this novel electrical power generation technology are projected to be less than \$0.30/W.

Keywords: Multiple Quantum Wells, Thermophotovoltaics, IV-VI Semiconductors, Thermal Energy Grid Storage

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1. INTRODUCTION

Costs for both solar and wind power generation systems are now falling below the costs for building power generation systems based on hydrocarbon combustion. These non-polluting energy sources, however, provide variable power. Without an energy storage system, they are projected to penetrate less than 15% of the total power grid [1]. Development of low cost and reliable energy storage systems for integration with solar and wind power sources will lead directly to complete decarbonization of the power grid. Recently, a low-cost thermal energy grid storage (TEGS) system has been proposed where solid state thermophotovoltaic (TPV) devices deliver electrical power from stored thermal energy [2]. Low energy storage costs, about 80% lower than Li-ion batteries, have been predicted for TPV-based TEGS systems [3]. In this paper, we describe a IV-VI semiconductor TPV device design for use in TEGS systems. Compared with III-V semiconductors, IV-VI semiconductor materials have the advantages of wider coverage of the mid-IR spectral region and a low cost, silicon-based materials growth technology. We model the performance parameters for two different IV-VI semiconductor multiple quantum well (MQW) materials designs. Results show that TPV devices made from these materials can generate grid-level power from stored thermal energy. We conclude that future work in this area will result in the development of low cost TPV cells for thermal energy conversion and speed the complete elimination of carbon dioxide pollution from electrical power generation systems.

IV-VI semiconductors, namely PbSe and related alloys, are attractive materials for TPV device fabrication. Their bandgaps can be varied from 0 eV to more than 1 eV. By contrast, III-V semiconductor materials cannot provide this materials design flexibility. Due to fundamental phase instabilities with high antimony content alloys, their bandgaps can only be as small as about 0.53 eV. Recently, multiple quantum well (MQW) materials based on PbSnSe wells and PbSrSe barriers with intersubband gap energies of 0.34 eV and 0.45 eV have been proposed for fabrication of dual junction TPV devices [4]. Bandgaps are determined by PbSnSe well width, so multi-junction TPV device materials can be grown and optimized by adjusting molecular beam epitaxy (MBE) shutter open times during quantum well growth. In addition to this manufacturing-friendly feature, these IV-VI semiconductor materials can be grown on low cost silicon substrates [5-12] and fabricated into devices using flip-chip epilayer transfer to copper for nearly optimal thin film device thermal management [13-15]. This semiconductor materials technology is well developed and currently at a stage where large-scale production can occur quickly. Further development of TPV devices based on these materials can result in low cost components for TEGS systems.

TEGS systems offer simple operation, high reliability, and low cost. Efficient TEGS systems use radiative heat transfer to store energy in a material’s phase transformation. Silicon is an attractive liquid-to-solid phase change energy storage material. It has a high enthalpy of fusion (latent heat) of 1.8 MJ/kg, which is about 4x larger than the energy storage density of Li-ion batteries. A molten silicon TEGS system can maintain steady state thermal radiator temperatures of 1364°C, about 50°C below silicon’s liquidus temperature. The radiant flux from a 1364°C thermal radiator such as SiC with an emissivity of 0.9 is shown in Figure 1. Indicated in the figure are two spectral regions, a bottom junction region covered by a MQW with a 343 meV bandgap and a top junction region covered by a material with a 450 meV bandgap. Both junctions are assumed to be thicker than 2 microns, a sufficient thickness to absorb more than 98% of non-reflected incident radiation based on the $>10^4\text{ cm}^{-1}$ absorption coefficient for IV-VI semiconductor materials. Figure 2 depicts a dual junction TPV device illuminated by a thermal radiator heated by molten silicon. With the top junction acting as a low pass filter, the bottom junction will be illuminated with a narrow and well-defined spectral band centered at a wavelength of about 3.1 μm . This allows implementation of an effective anti-reflective coating to increase optical absorption for the bottom junction. Our model uses 80% as the portion of thermal radiation within the 343 meV to 450 meV band that is absorbed by the bottom junction and 65% as the portion of thermal radiation within the 450 meV to 800 meV band that is absorbed by the top junction. Table I lists the incident optical power in the top and bottom junction bands and the estimated charge carrier generation rates for each junction based on assumed average photon energies and the amount of absorption by each junction.

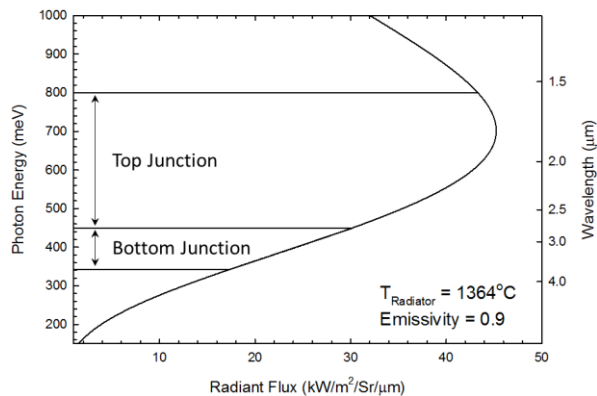


Figure 1: Photon energy versus radiant flux from a 1364°C thermal radiator with an emissivity of 0.9. The spectral coverage by a dual junction TPV device composed of IV-VI semiconductor MQW materials is indicated.

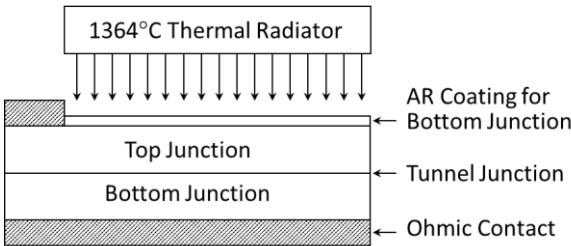


Figure 2: Dual junction TPV device with illumination from a thermal radiator heated by molten silicon. An anti-reflective coating for high transmission of ~3.1 μm photons to the bottom junction is indicated.

Table I: Parameters for determining charge carrier generation rates in the top and bottom junctions of a dual junction TPV device made from IV-VI semiconductor MQW materials for a 1364°C thermal radiator with an emissivity of 0.9.

$T_{\text{Radiator}} = 1364^{\circ}\text{C}$	Band (meV)	Incident Power (W/cm^2)	Average Photon Energy (meV)	Absorption (% of band)	Generation Rate ($\text{s}^{-1}\text{cm}^{-2}$)
Top Junction	450-800	14.6	625	65	9.48×10^{19}
Bottom Junction	343-450	6.02	396	80	7.61×10^{19}

2. TPV MATERIALS AND DEVICE MODEL

IV-VI semiconductor MQW materials composed of alternating layers of $\text{Pb}_{0.80}\text{Sr}_{0.20}\text{Se}$ ($E_g = 840$ meV) and $\text{Pb}_{0.81}\text{Sn}_{0.19}\text{Se}$ ($E_g = 84$ meV) were recently proposed as useful for TPV device fabrication [4]. Four-fold L-valley band degeneracy is removed when these materials are grown on (111)-oriented substrates. Quantum confinement in the [111] direction creates three-fold degenerate oblique valleys with relatively small effective masses and a single normal valley with a larger effective mass. Schrödinger's equation and band structure parameters for PbSnSe, i.e. longitudinal and transverse effective masses for L-valley electrons and holes, were used to show that well widths of 4.9 nm and 3.5 nm give bandgaps of 343 meV and 450 meV, respectively. Compared to bulk IV-VI semiconductor alloys with the same bandgap energies, the effective densities of states in the conduction and valence bands for these MQW materials are about a factor of three lower. Thermal generation of intrinsic charge carriers is thus reduced by using this quantum confinement effect. Smaller intrinsic carrier densities cause smaller reverse bias saturation currents under dark conditions and larger open circuit voltages under thermal illumination conditions. These two MQW materials can be used in a dual junction TPV device that will absorb the thermal radiation shown in Figure 1. Table II provides the intrinsic charge carrier densities, calculated using the procedures outlined in ref. [4], for these two materials at three different temperatures.

Table II: Intrinsic charge carrier densities for 450 meV and 343 meV bandgap PbSnSe/PbSrSe MQW materials. The three different temperatures span a range that can be expected for different levels of TPV module cooling in a TEGS system.

PbSnSe/PbSrSe MQW Material		Intrinsic Carrier Density, n_i (cm^{-3})		
Well Width (nm)	E_g (meV)	7°C	23°C	50°C
3.5	450	7.36×10^{13}	1.51×10^{14}	3.11×10^{14}
4.9	343	4.92×10^{14}	8.71×10^{14}	1.55×10^{15}

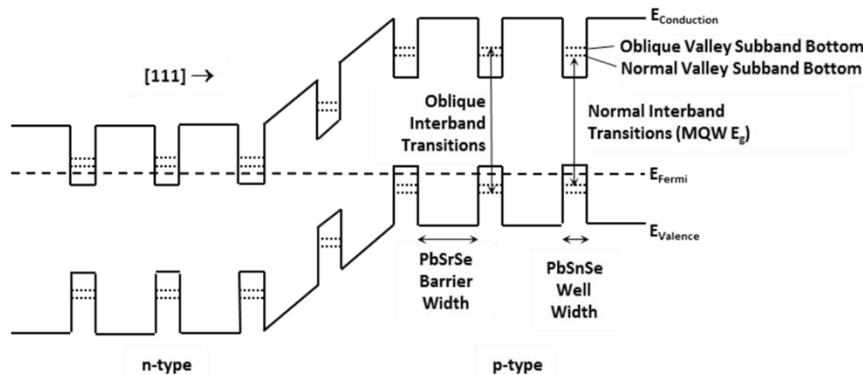


Figure 3: Band diagram depiction of a pn junction fabricated with IV-VI semiconductor MQW materials. One dimensional confinement of charge carriers in (111)-oriented PbSnSe quantum wells removes L-valley degeneracy. This decreases effective densities of states and increases open circuit voltage [4].

Figure 3 shows an energy level diagram for a IV-VI semiconductor pn junction made from a MQW material. The bandgap of the MQW material is the energy spacing between the edges of the normal valley subbands, which are determined by PbSnSe well width. Two of these pn junctions with different PbSnSe well widths, connected in series with a heavily-doped pn tunnel junction, will comprise a dual junction TPV device. For doping levels of $3 \times 10^{17} \text{ cm}^{-3}$ on the n-type and p-type sides for the 450 meV bandgap top junction the built-in potential barrier and depletion width are 390 meV and 81 nm, respectively. For the 343 meV bandgap bottom junction, the built-in potential barrier and depletion width are 300 meV and 71 nm, respectively. These values change only slightly, less than few percent, over the 7°C to 50°C temperature range considered here. Equation (1) expresses reverse bias saturation current in terms of intrinsic carrier density, majority carrier concentration, temperature, charge carrier mobility, and minority carrier lifetime. Charge carrier mobilities have been measured in IV-VI semiconductor layers grown on silicon substrates, and they range from about 300 cm^2/Vs for high strontium content PbSrSe alloys to more than 1,500 cm^2/Vs for high tin content PbSnSe alloys. Decreasing

values from 700 cm²/Vs to 400 cm²/Vs as temperature increases for the 450 meV MQW material, and 1000 cm²/Vs to 700 cm²/Vs for the higher tin content 343 meV MQW material are assumed here. Based on measurements of PbS/PbSrS heterojunctions grown by MBE on Si [16], a minority carrier lifetime of 10 ns for both electrons and holes is assumed. Table III shows calculated J_o values for both MQW materials at different temperatures. Minority carrier diffusion lengths are also provided.

$$J_o = n_i^2 \sqrt{\frac{qkT\mu}{\tau}} \left(\frac{1}{n_n} + \frac{1}{p_p} \right) \quad (1)$$

Table III: Mobilities, lifetimes, and diffusion lengths for electrons and holes, and reverse bias saturation currents for 450 meV and 343 meV bandgap MQW materials.

T (°C)	E _g (meV)	μ _n , μ _p (cm ² /Vs)	τ _n , τ _p (ns)	L _n , L _p (μm)	J _o (mA/cm ²)
7°C	450	700	10	4.1	0.002
	343	1000	10	4.9	0.127
23°C	450	550	10	3.7	0.009
	343	850	10	4.7	0.377
50°C	450	400	10	3.3	0.034
	343	700	10	4.4	1.132

Short circuit current density is determined by electronic charge, generation rate, and internal quantum efficiency (IQE) as shown in equation (2).

$$J_{sc} = qG \times IQE \quad (2)$$

Generation rates are given in Table I. IQE is the percentage of photogenerated charge carriers in the semiconductor that reach the junction and are collected by an ohmic contact. The long diffusion lengths of electrons and holes, about 4 times larger than the ~1 μm thicknesses of the n-type and p-type layers that comprise a dual junction TPV device, will give IQE values close to 100% for the bottom junction. The top junction, where more hot electron generation occurs, is assumed to have an IQE of 80%. In this case, both junctions will have a short circuit current value of 12.1 A/cm². This value is used in Table IV to calculate open circuit voltage, fill factor [17], and electrical power generation rate as shown in equations (3-5). Figure 4 shows total calculated electrical power generation density for this dual junction TPV device along with the contribution by each junction as a function of temperature.

$$V_{oc} = \frac{kT}{q} \ln \left(\frac{J_{sc}}{J_o} + 1 \right) \quad (3)$$

$$FF = \frac{V_{oc} - \ln(V_{oc} + 0.72)}{V_{oc} + 1} \quad (4)$$

$$P_{el} = FF \times V_{oc} \times J_{sc} \quad (5)$$

Table IV: Short circuit current density, open circuit voltage, fill factor and electron and hole mobilities, lifetimes, diffusion lengths, and electrical power generation density for 450 meV and 343 meV bandgap MQW materials at three temperatures.

T (°C)	Junction	E _g (meV)	J _{sc} (A/cm ²)	V _{oc} (mV)	FF (%)	P _{el} (W/cm ²)
7°C	Top	450	12.1	262	70	2.20
	Bottom	343	12.1	166	60	1.20
23°C	Top	450	12.1	242	68	1.99
	Bottom	343	12.1	147	57	1.02
50°C	Top	450	12.1	227	67	1.84
	Bottom	343	12.1	130	54	0.86

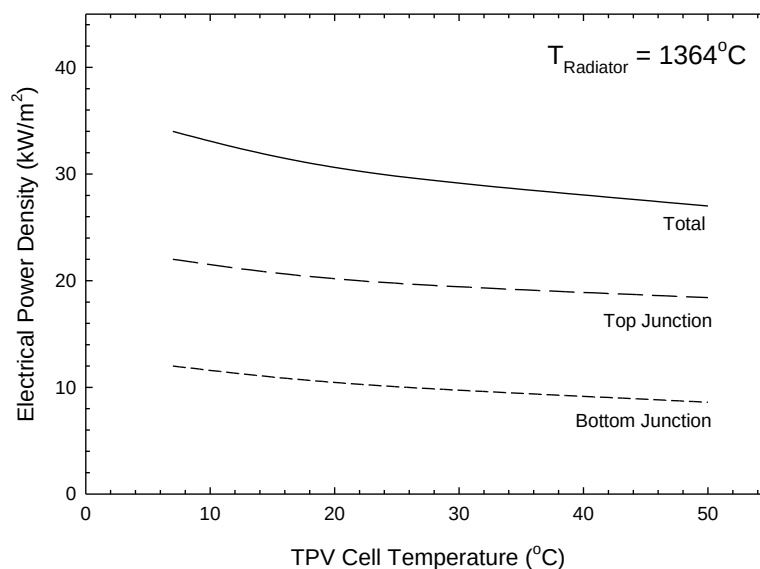


Figure 4: Dual junction IV-VI semiconductor MQW TPV device electrical power generation density as a function of temperature for a thermal radiator temperature of 1364°C. Total power and the contribution from each junction are plotted.

3. DISCUSSION AND SUMMARY

A thermal photovoltaic device technology based on narrow bandgap IV-VI semiconductor MQW materials has been described. Electrical power densities for a dual junction device as a function of operation temperature have been predicted for a thermal radiator temperature of 1364°C, see Figure 4. All assumptions made in our model, including minority carrier lifetimes for photogenerated charge carriers, are supported by experimental data obtained from narrow bandgap IV-VI semiconductor layers grown on silicon substrates. The dual junction TPV device modeled here can serve as a benchmark design for development of future TPV devices with high power conversion efficiencies. For example, it is possible to grow MQW structures with three different bandgaps for fabrication of triple junction TPV devices. An outline for optimal designs of multi-junction TPV devices has recently been published, and triple junction TPV devices made from materials with bandgaps below 500 meV were shown to have the potential for power conversion efficiencies greater than 55% for a 1400°C radiator temperature [18]. Further work in this area can help to build a foundation for the development of practical TPV devices for use in thermal energy grid storage systems.

The availability of low cost solar PV power makes the use of TPV devices in a TEGS system economically interesting. For example, it is likely that utility-scale solar PV projects will have costs below \$0.50/W in about 3-5 years. This is based on the recent completion of the 1.17 GW Noor Abu Dhabi solar PV power plant in the UAE, which had a cost of \$0.77/W, a decrease from the \$0.95/W cost a few years ago for utility-scale projects. If a TEGS-TPV energy storage system has a round trip efficiency of 50%, then a solar PV system oversized by a factor of three relative to the average daytime peak load can store about 10 hours of grid load power on a daily basis. The costs for a complete TEGS-TPV system are estimated to be about \$0.30/W when multi-junction TPV cells with power conversion efficiencies of 50% are used [3]. Therefore, the total cost of a combined solar PV and TEGS-TPV system, which can deliver consistent and reliable grid power, is estimated to be about \$1.80/W. This is on par with the current cost for building a gas turbine power plant. A solar PV/TEGS-TPV system, due to overall mechanical and chemical simplicity, will be expected to have a long service lifetime, up to 40 years for many of its core components, and it will have no fuel costs. The levelized cost of electricity provided by such a clean energy system will most likely be lower than any other form of grid power technology.

Future work on developing efficient TPV materials and device technologies can follow the same approach as recent work on developing multi-junction materials and device technologies for solar PV applications. In the TPV case, however, the potential positive impact can be much greater. By enabling a low cost way to store grid-level energy, an effective TPV device technology will expand the penetration of solar PV systems in power grids significantly. Work in this area offers a technologically and economically viable solution for complete decarbonization of the Earth's power grids.

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